

Taenia solium

Miguel Perez

Taxonomy

- Domain: Eukaryota
- Kingdom: Animalia
- Phylum: Platyhelminthes
- Subclass: Cestoda
- Order: Cyclophyllidea
- Family: Taeniidae
- Genus: Taenia
- Species: solium

T. solium



Img source:

<https://www.nikonsmallworld.com/galleries/2017-photomicrography-competition/taenia-solium-everted-scolex>

Historical Facts!

- Ancient Disease!
 - Tapeworms in ancient Egyptian mummies [3000-1500 BC]
- Hipólito Unanue [1792]
 - First description of simultaneous taeniasis and cysticercosis





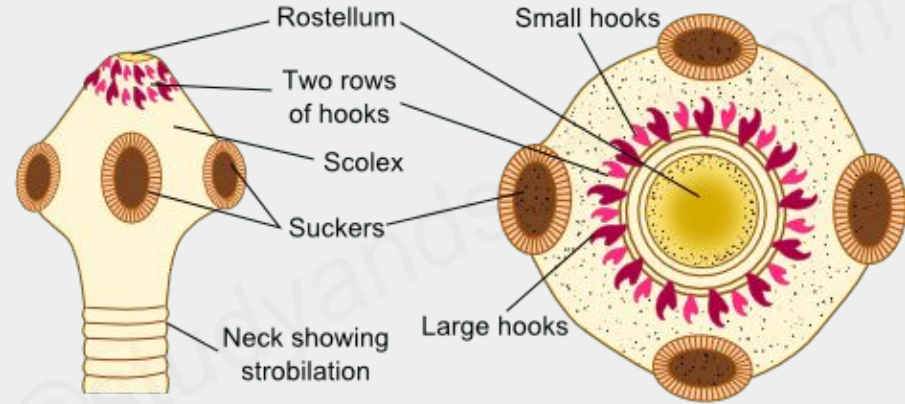
- Friedrich Küchenmeister [19th Century]
 - Demonstration of human intestinal taeniasis via cysticercus



- Kozen Yoshino [1930's]
 - Confirmed cysticercosis life cycle → tapeworm in humans through group and self experimentation

Morphology

- Mature tapeworm key features:
 - Armed scolex with rostellum
 - 2 circles, 22-32 hooks (alternating)
 - Spheroid scolex

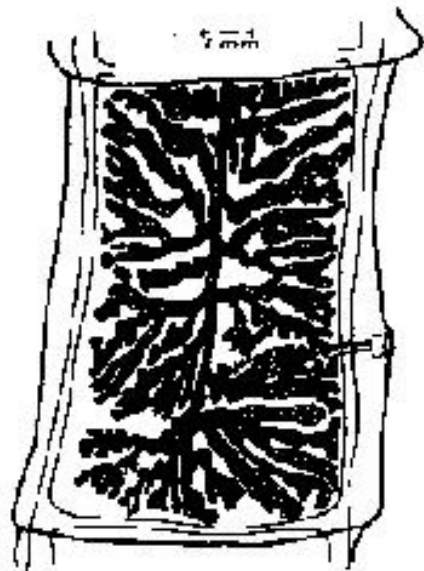


TAENIA SOLIUM - FRONT AND TOP VIEW

©studyandscore.com

- Proglottids
 - Gravid: 7-11 uterine branches
 - Key identifying feature

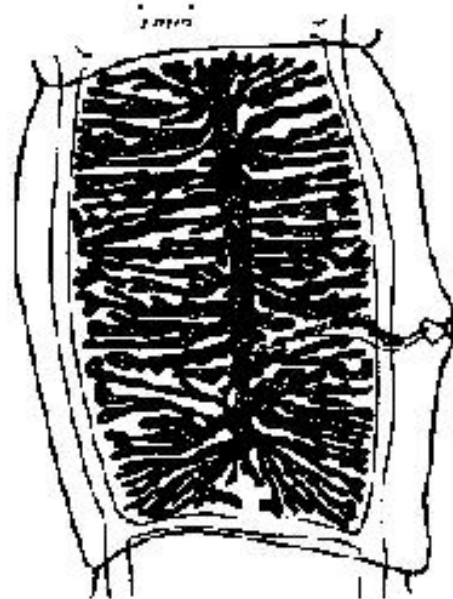
UTERINE BRANCH COMPARISON



GRAVID PROGLOTTID
T. solium



EGG



GRAVID PROGLOTTID
T. saginata

Comparing the two *Taenia* species

Characteristic	<i>Taenia saginata</i>	<i>Taenia solium</i>
Intermediate Host	Cattle, reindeer	Pig, wild boar
Site of Development	Muscle, viscera	Brain, skin, muscle
Scolex: adult worm	No hooks	Hooks
Scolex: cysticercus	No rostellum	Rostellum & hooks
Proglottids: uterine branches	23 (14 – 32) *	8 (7 – 11) *
Passing of proglottids	Single, spontaneous	In groups, passively
Ovary	Two lobes	Three lobes
Vagina: sphincter muscle	Present	Absent

2 Life Cycle Paths:

Solium taeniasis

- **Ingestion of cysticerci**
 - Intermediate host: pig
- Bladder worm attaches to small intestine wall
- Maturation!
 - Within 2-3 months
 - Lifespan: ~25 years
- Gravid proglottids released
 - Groups of 5-6, passively
- Humans are definitive host

Human cysticercosis

- **Ingestion of eggs/proglottids**
 - Infected human fecal matter
 - Infect others or autoinfection
- Lower intestine → stomach or duodenum
- Travel through blood/lymph
- No tissue/organ is safe
 - Common sites: subcutaneous connective tissues, eyes, brain, muscles, heart, liver, lungs, and coelom.

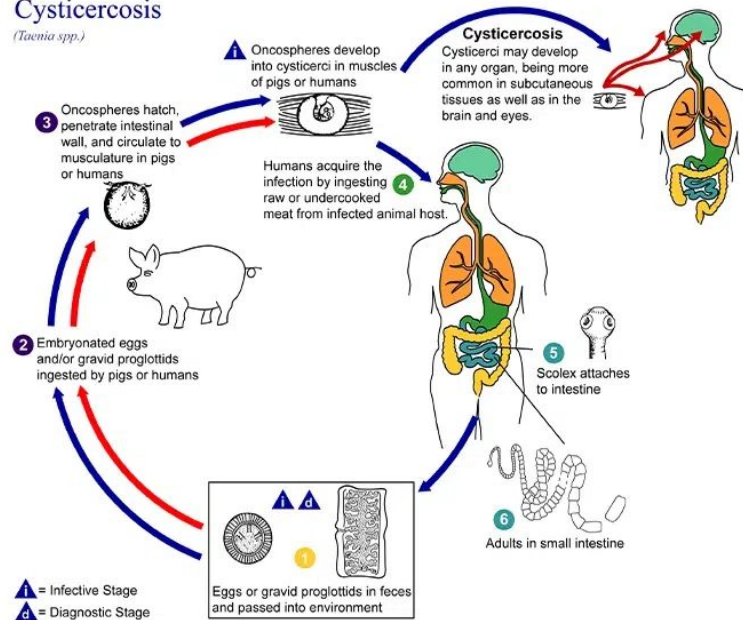
TAPEWORM LIFE CYCLE



Img
source:
<https://www.geeksforgeeks.org/tape-worm-life-cycle/>

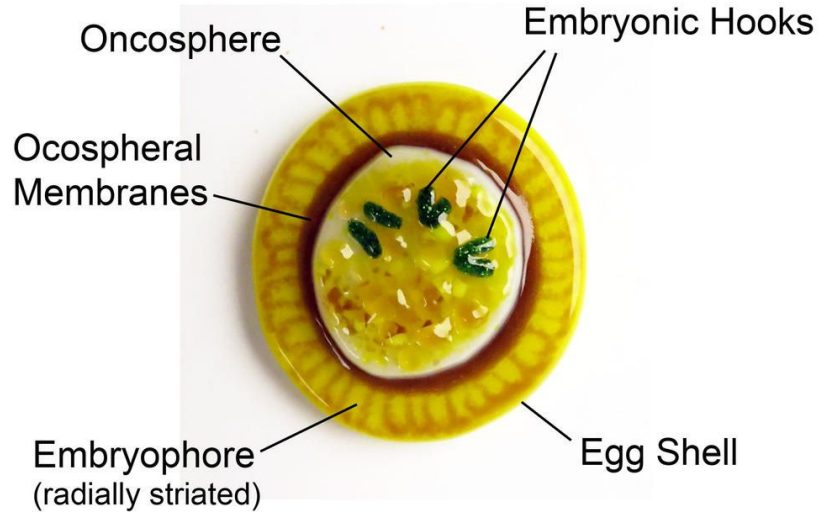
Cysticercosis

(*Taenia spp.*)



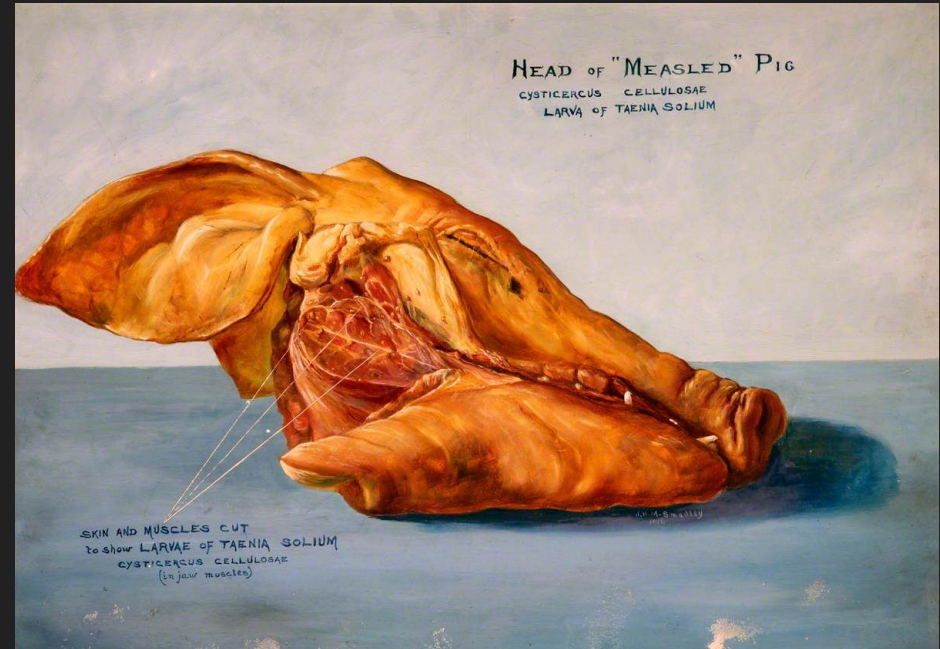
- Remember: eat cysticerci → get solium taeniasis; eat eggs/proglottids → get cysticercosis

Taenia solium Tapeworm Egg



Img source:

<https://www.deviantart.com/trilobiteglassworks/art/Taenia-solium-Tapeworm-Egg-Fused-Glass-791078227>



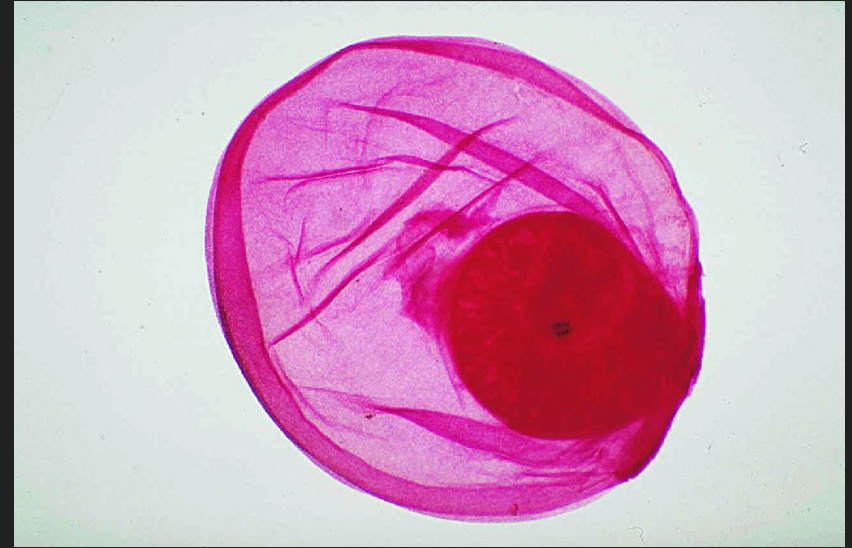
Img source:

<https://artuk.org/discover/artworks/head-of-measled-pig-cysticercus-cellulosae-larva-of-tapeworm-taenia-solium-104373>



Img source: <https://www.cdc.gov/dpdx/taeniasis/index.html>

- Note the spheroid scolex



- Fluid-filled cysticercus
 - Note the developing hooks
 - Can be in either pigs or humans

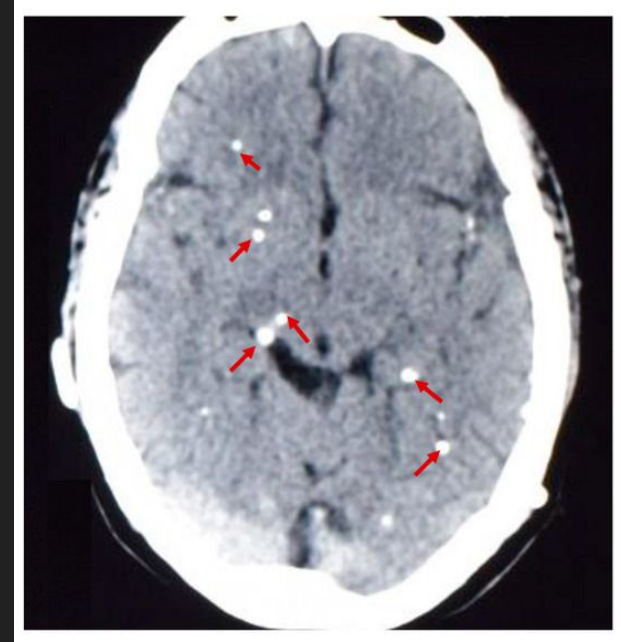
Giant swelling pseudotumor

- Fluid build up in
cysticercus
 - Can have
liters
- Connective
tissue forming
around cyst



Continuation of human cysticercosis

- Death of cysticerci → calcification
 - Severe inflammatory response
 - 1 year = asymptomatic
 - Many years = serious symptoms
 - Brain infection and calcification
 - Neurocysticercosis
 - Content warning next slide



Img source:

<https://www.mdpi.com/2076-0817/13/1/26>



Monsters Inside Me

<https://www.youtube.com/watch?v=qlTTyuvA5wQ>

Epidemiology

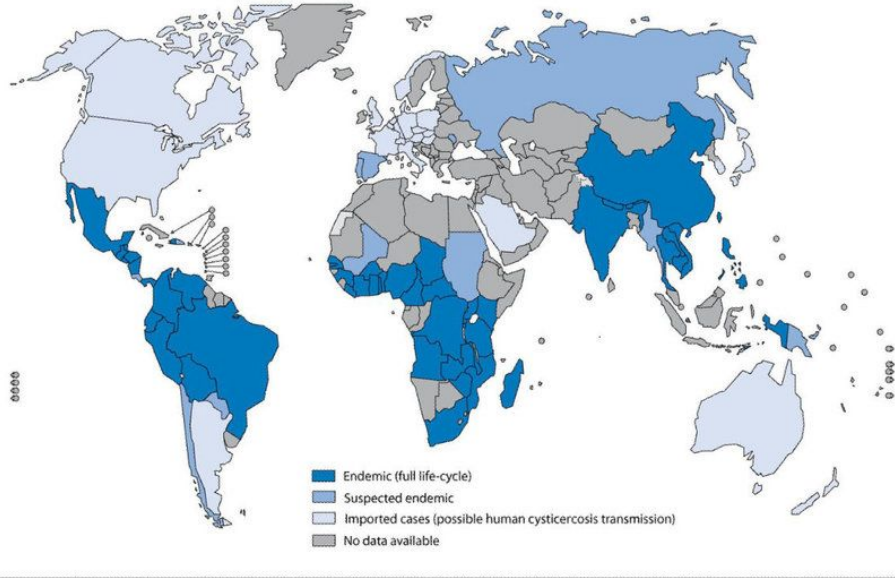
Areas of common infection:

- Africa, India, China, South and Central America, Mexico

Impact of culture and diet on infection:

- Muslim and Jewish community
- Vegetarians

Countries and areas at risk of cysticercosis, 2009



The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted lines on maps represent approximate border lines for which there may not yet be full agreement. © WHO 2010. All rights reserved

Data Source: World Health Organization
Map Production: Control of Neglected
Tropical Diseases (NTD)
World Health Organization



Img source:

https://www.researchgate.net/figure/World-map-showing-the-distribution-of-Taenia-solium-taeniasis-cysticercosis-transmission_fig5_51632511

Important Aspect

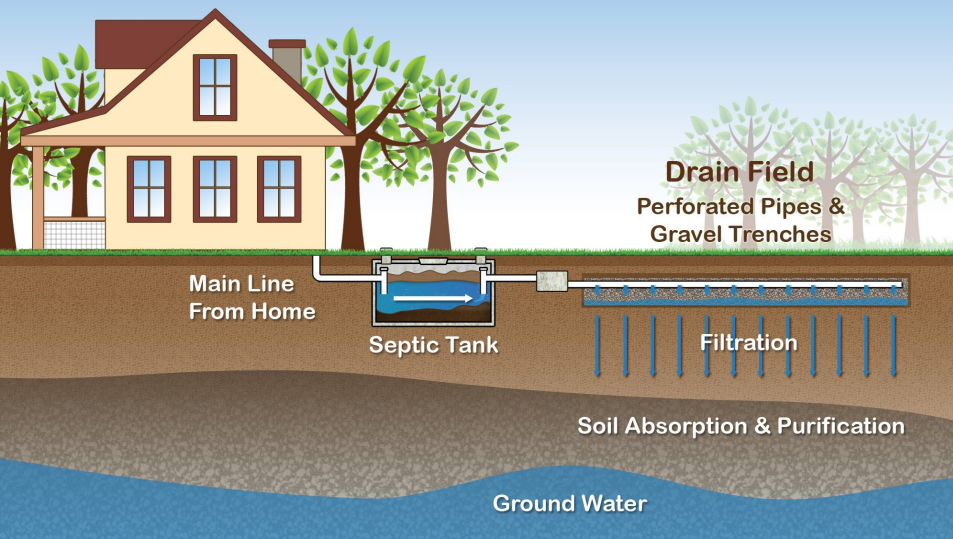
Cysticercosis could happen to anyone

- Vegetarian or non-vegetarian
- Tainted salads or water
- Common in same places as taeniasis



Img source: <https://ijdv1.com/extensive-disseminated-cysticercosis/>

Conventional Septic System

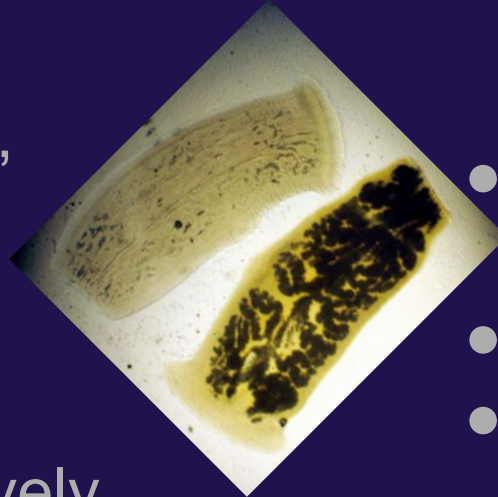


- Common in unsanitary conditions
 - Pigs eating eggs from human feces
 - Flooded pig farms
 - Ex: broken septic tank

Pathogenesis

Solium taeniasis

- Non-specific symptoms
 - Vague abdominal pain
 - Armed scolex
 - Nausea, Diarrhea, Constipation
- Gravid proglottids in stool
 - 5-6 groups, passively



Cysticercosis

- May remain asymptomatic for many years
- Muscle spasms, weakness, general weakness
- Visible/palpable subcutaneous nodules
- Neurocysticercosis
- Ocular cysticercosis

Treatment, Prevention, and Control

- Praziquantel or Albendazole [1-2 weeks]
 - For both solium taeniasis and cysticercosis
 - 5-10 mg/kg Praziquantel
 - 400 mg Albendazole
- Surgery for cysticerci
 - Meat inspection
 - Properly cooking pork
 - Safe disposal of human waste
 - Proper washing of vegetables
 - Sexual habits and proper hygiene
 - Boiling and filtering of water
 - Raw fruits and vegetables in endemic regions





Case Study 1 - Neurocysticercosis

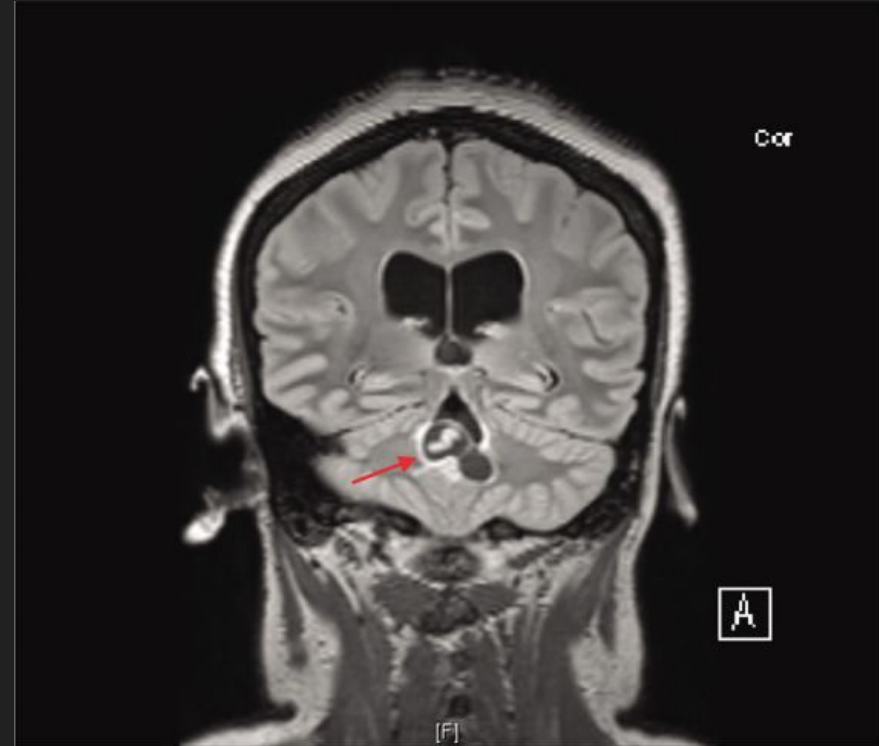
- 36-year old female
- History of adult onset seizures
 - Born and raised in Mexico
 - Moved to US at age 8
 - First seizure: 10 years prior
- Denied report of any intracranial lesions

Further examination

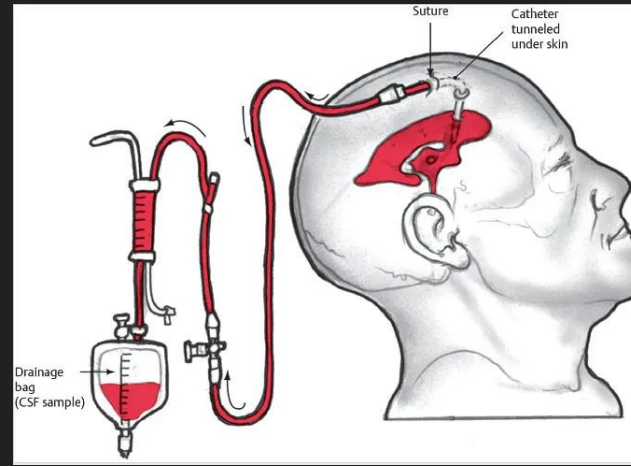
- Worsening headaches
 - Worse at night and in the morning, but better throughout day
- CT scan
 - Ventriculomegaly
 - Cerebrospinal fluid (CSF) buildup in brain ventricles
 - Transependymal flow
 - CSF in ventricles → brain tissue
 - Obstruction at fourth ventricle level
- No neurological deficits found

Further examination pt. 2

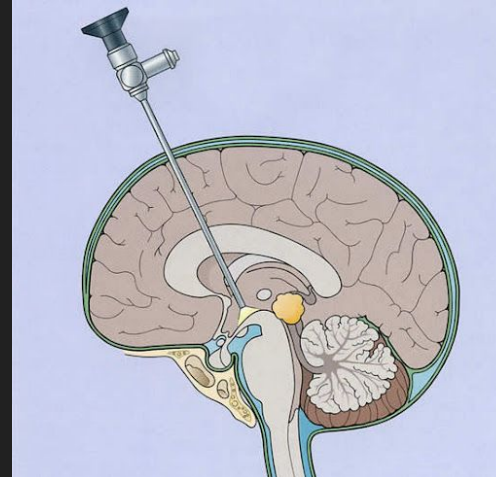
- MRI
 - Lobulated cystic mass within fourth ventricle
 - Internal linear enhancement
 - Increased brightness in area
 - Indicative of neurocysticercosis
 - Worm within cyst



- Elective surgery scheduled
- Right frontal burr hole (hole in skull)
 - External Ventricular Drain (EVD) for CSF draining
 - Endoscopic Third Ventriculostomy (ETV) = bypass for CSF to travel through to normal pathway

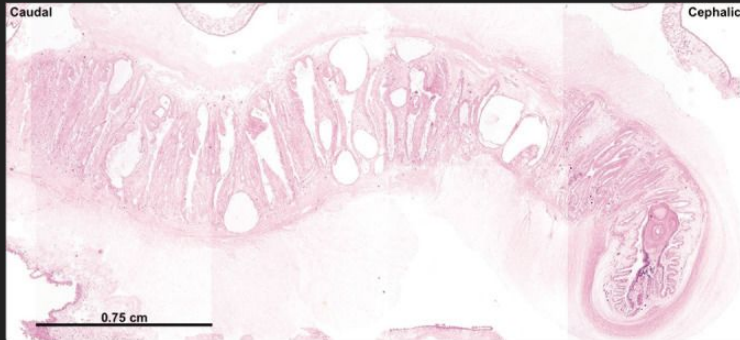


Img source:
<https://neupsykey.com/external-ventricular-drain/>



Img source:
http://oxfordneurosurgeon.com/treatments/endoscopic-surgery_intraventricular-endoscopy

- Suboccipital craniotomy
 - Removal of skull at base of the head
 - Access to back of the brain
- Removal of cystic mass *en bloc* (in one piece)



- Identification of mature *T. solium* plerocercoid larvae (1.2 X 0.5 X 2 cm)

Post Op

- No neurological deficits
- EVD removed on day 2
- Ophthalmologic exam
 - Concern for chronic, inactive choroidal inflammatory process
- Suspected of auto-infection
 - History of seizures
 - Recent presentation with active lesion



- Albendazole treatment (21 days)
- Dexamethasone taper (14 days)
 - Guard against aseptic meningitis
- Single dose of Praziquantel
 - Gastrointestinal auto-infection risk reduction



Case Study 2: Biliary solium taeniasis

- 70-year old Korean man
- Recurrent vomiting and colicky pain
- Fever (38.0°C), physical exam conducted
- Showed positivity for Murphy's sign
 - Gallbladder inflammation



- Denied history of:
 - Hepatobiliary disease (bile ducts, liver, gallbladder)
 - Parasitic infections
 - Digestive problems
 - Passing of parasites in feces
 - Eating uncooked food

Inflammatory
response

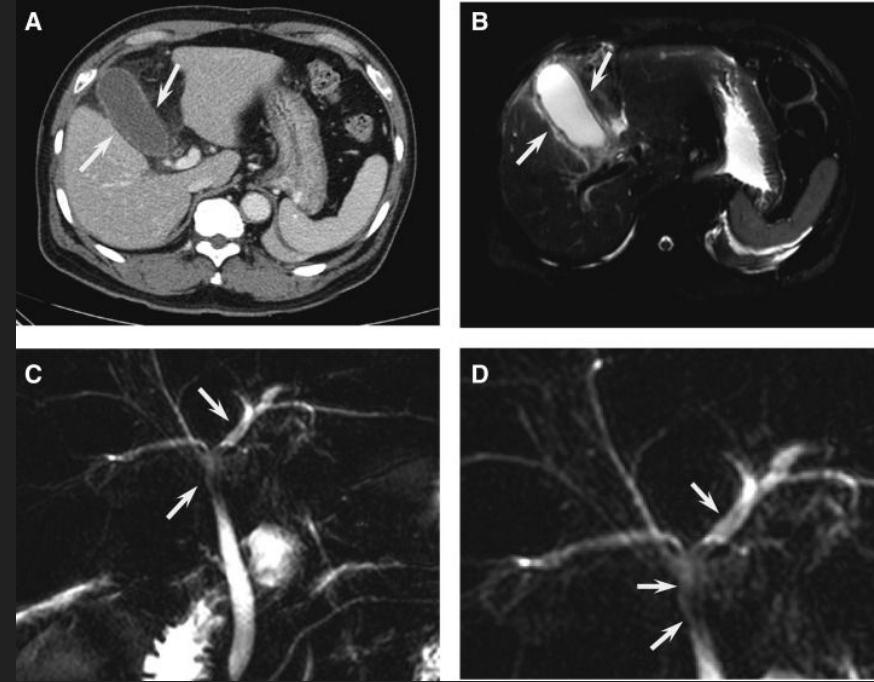
Severe inflammatory
status

	Patient Readings	Normal Readings
WBC count	20.6 * 10 ⁹ /L	4.5 - 11.0 * 10 ⁹ /L
LDH serum levels	275 IU/L	125 - 220 IU/L
CRP serum levels	26.84 mg/dL	0.3 - 1.0 mg/dL
Fasting glucose serum levels	169 mg/dL	70 - 100 mg/dL
HbA1c range	7.60%	6.50%

Diabetes mellitus
- Hyperglycemia

CT Scan

- Thickened gallbladder wall
 - Suggests tissue death of wall due to reduced blood flow
 - Tissue swelling around gallbladder → inflammation

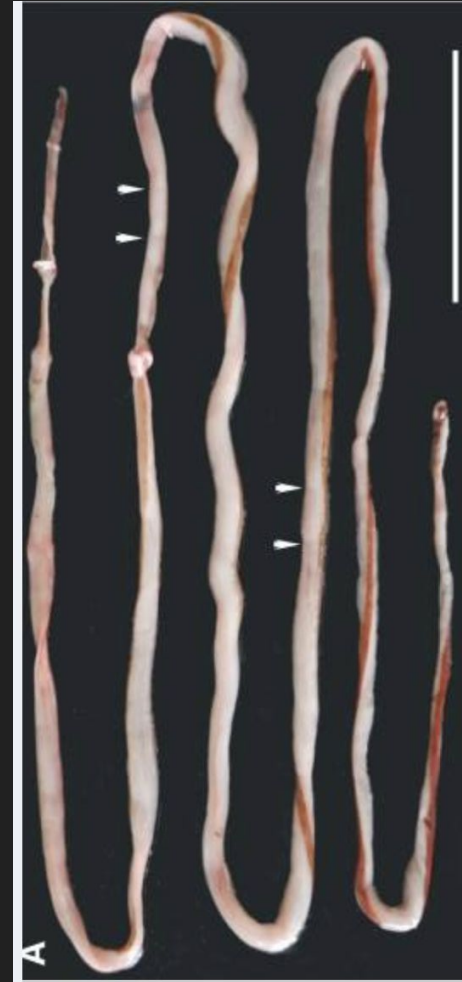
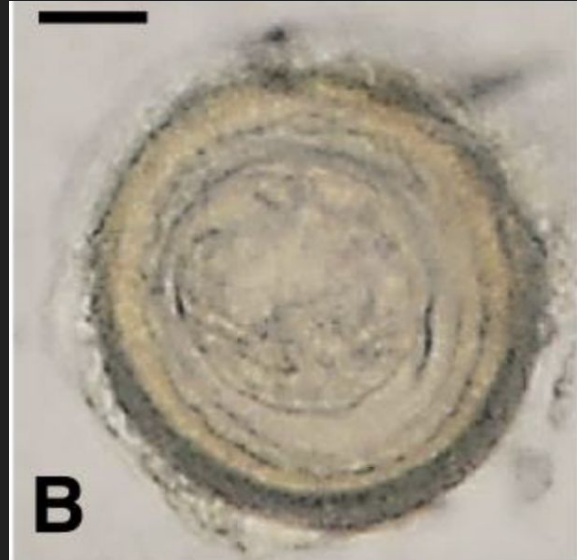
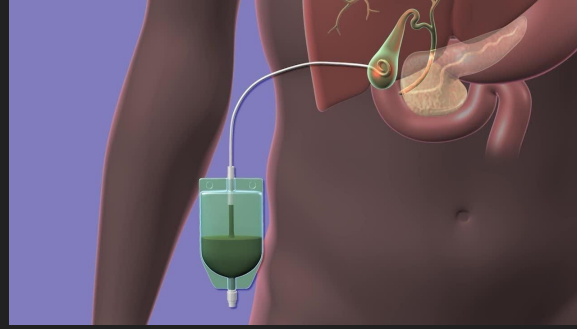


MRCP Image

- Darkened linear structures within intra- and extrahepatic bile ducts
 - Immediately prescribed ceftriaxone and metronidazole

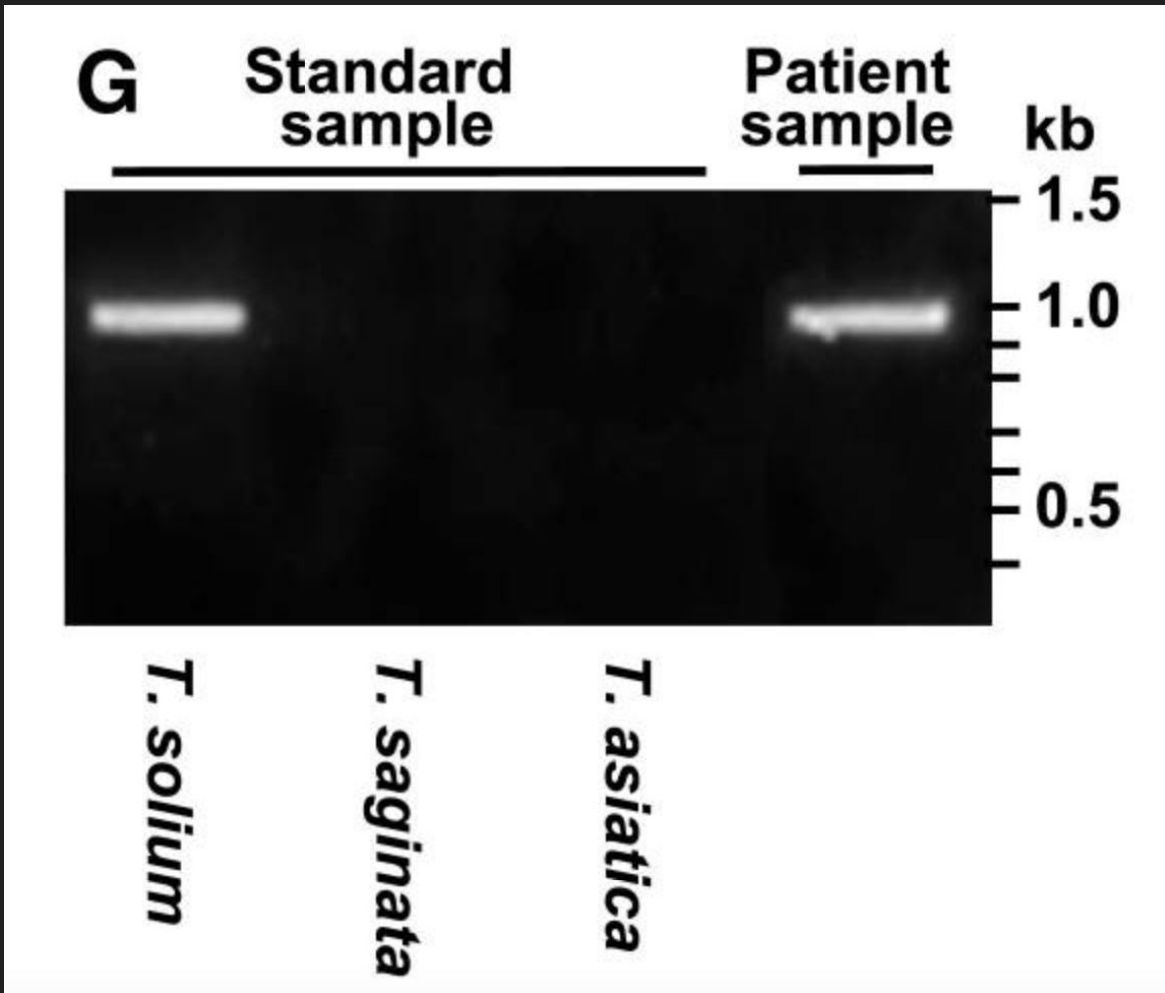
3rd day of admission

- Percutaneous transhepatic gallbladder drainage
- During fluid irrigation of catheter, long-bodied parasite was drawn out
- Highly degenerated proglottids were found on posterior end
- Striated embryophore with hooklets found



ID-ing the parasite

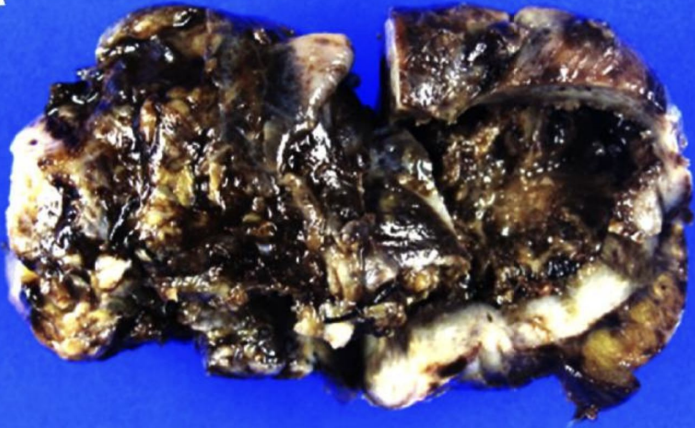
- PCR test conducted with worm DNA



Post Worm Removal

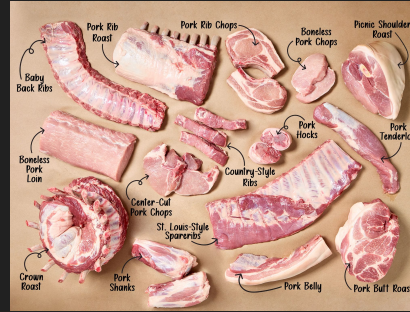
- Worm removal → gastrointestinal symptoms subsided
- Treated with single dose of Praziquantel (10 mg/kg)
- Persistent fever with inflammatory signs → inflamed gallbladder removal
 - Laparoscopic cholecystectomy
- Residual parasitic eggs/worms not observed
 - Recovery after 1 month
 - All serum levels returned to normal

A



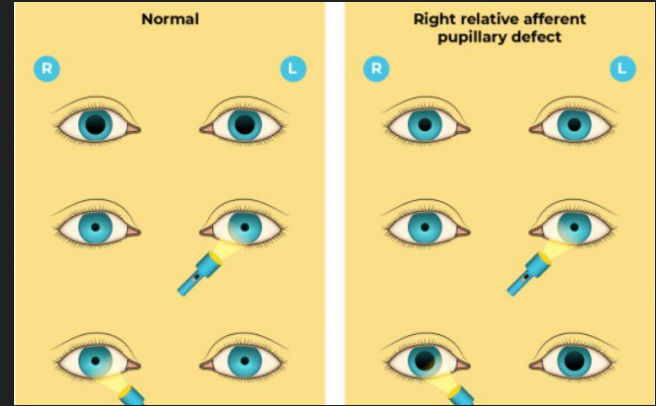
Case Study 3 - Subretinal Cysticercosis

- 42-year old female
- Decrease in right eye vision
 - 1 month
 - Painless, progressive
 - Associated with floaters
- 10 years without vision in left eye
- History of left-sided hemiplegia and Multiple Sclerosis for 10 years
- No history of Diabetes Mellitus or Hypertension
- Vegetarian diet for 2 years, ate pork meat 2 years prior



Eye tests and observations

- Best Corrected Visual Acuity (BCVA)
 - Right eye: 5/60
 - Left eye: light perception only
- Intraocular pressure levels were normal
 - Typical range of 10-20 mmHg
 - 11 mmHg for the right, 15 mmHg for the left
- Rapid Afferent Pupillary Defect in left eye
 - Irregular response to light stimuli

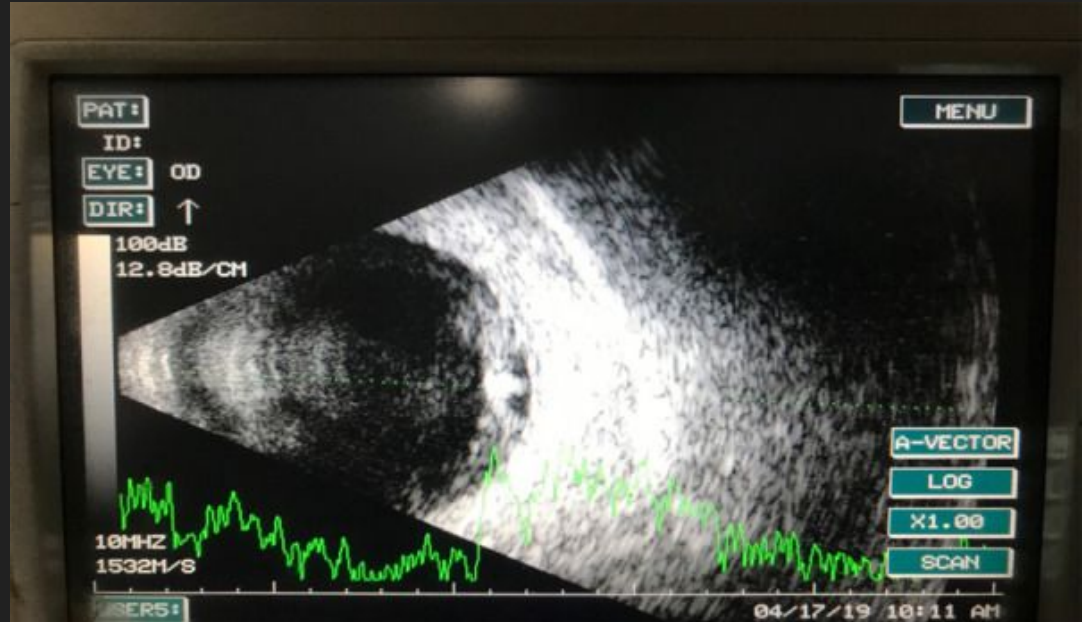


- Whitish round cystic mass found
 - Posterior vitreous attachment
 - Vitreous = referring to the jelly-like substance inside the eyeball



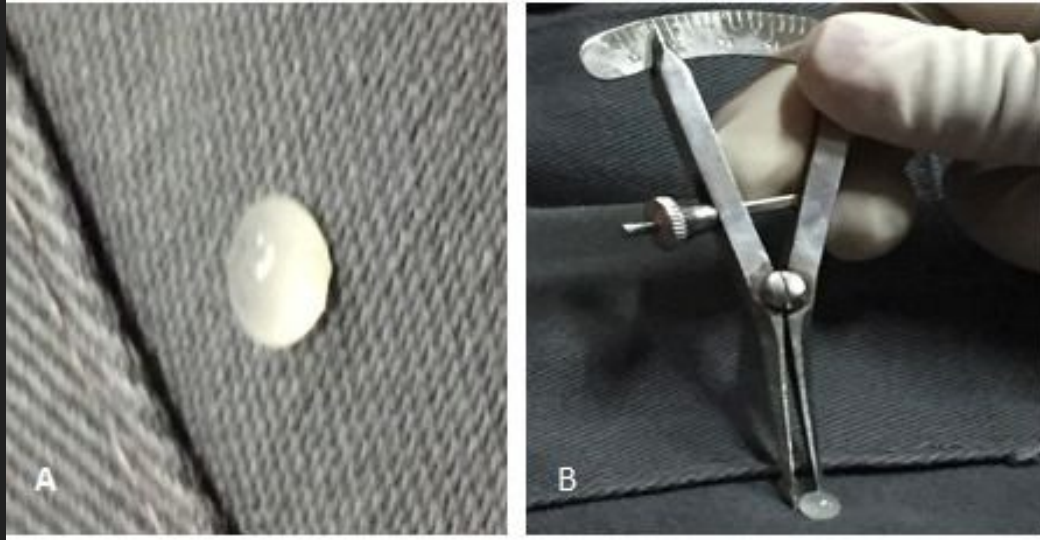
Further examination of the cyst

- Dense white material on cyst
 - Cysticercoid with scolex
- Brightness scan
 - Highly reflective elevation on the dome suggests scolex



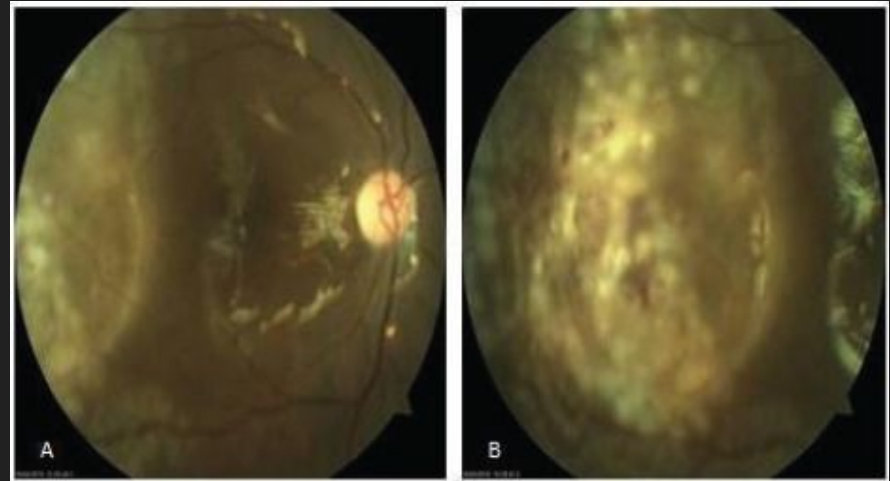
- MRI
 - Lesions on the brain
 - Chronic hemorrhagic lesions
 - Represents calcific foci (specific area of abnormality)
 - Possible calcified cysticerci
- Blood tests and serological tests were normal
- No parasites found in stool
- Provisional diagnosis:
 - Right eye subretinal cyst due to cysticercus
 - Left eye atrophy due to multiple sclerosis

Operation



- Pars plana vitrectomy of right eye
 - Incision in eye to access vitreous gel
- Subretinal cyst removed
 - Sent out for histopathological examination
- Gross examination
 - 5 mm cyst
 - Whitish material within cyst suggests cysticerci with scolex

- BCVA on first postoperative day
 - 1/60 for right eye
 - Intraocular pressure at 10 mmHg
- Fundus examination
 - Back of the eye
 - Well-attached retina
 - Laser marks
 - Silicone oil
 - Fill space of vitreous in post op



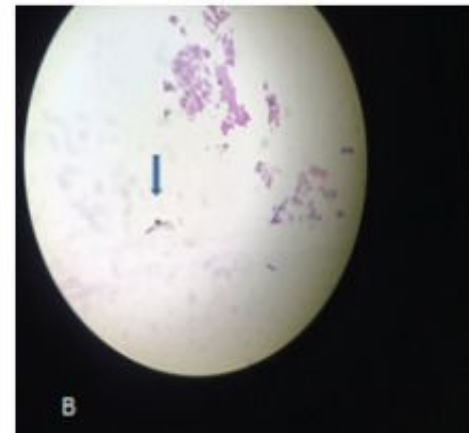
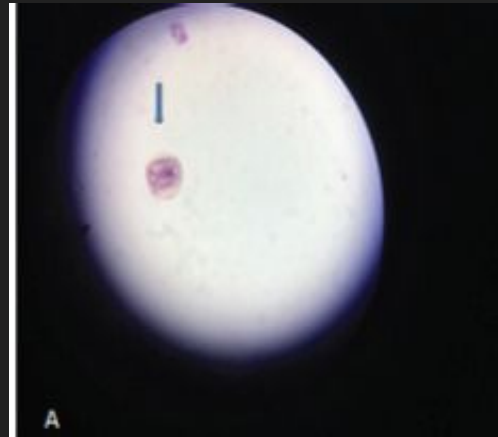
2 weeks later

- BCVA of 6/36 with +6.00 Dioptre spherical lens
 - To correct sight
- Fundus examination similar to before

Histopathology

- Presence of suckers and hooks of scolex
 - Diagnosis of cysticercosis ✓

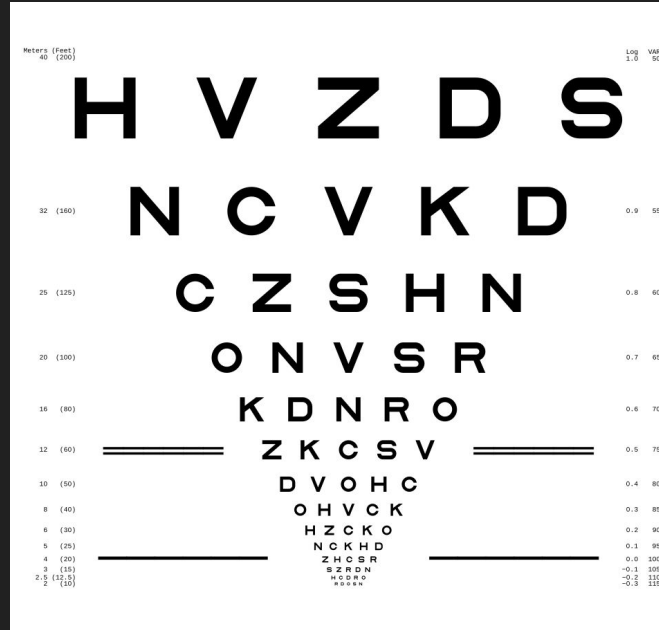
- Sucker on A, scolex on B



- Patient prescribed glasses
 - 6/36 BCVA
 - Over the course of 1 month

3 months later

- Silicone oil removed
- Final BCVA: 6/12



Question 1:

What was the key indicator of neurocysticercosis in case study 1?

MRI scan of the cyst with a worm in it

Question 2

Why was the taeniasis in case study 2 particularly bizarre?

Because the worm was found in the bile duct, not the small intestine

Works Cited

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